Nature Notes

Sprint 1

Team 5

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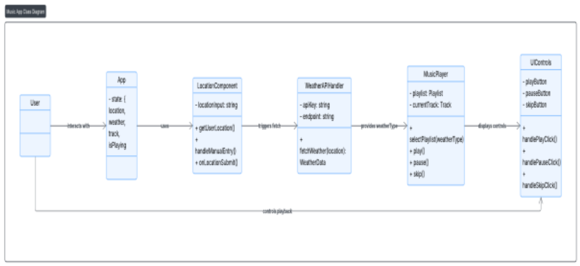
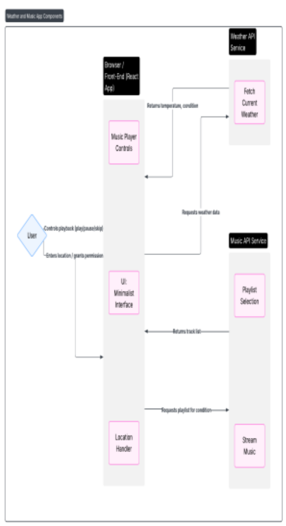
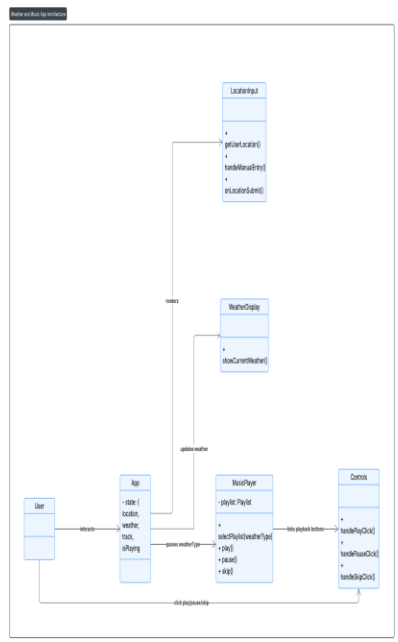
<u>**Architecture Description**</u>

Sprint 1 sees the building of the frontend architecture. We set up the React web framework. In addition, we added the weather API call. This sets the stage for a solid foundation before adding the backend work at sprint 2.







UML#1: Use Case Diagram

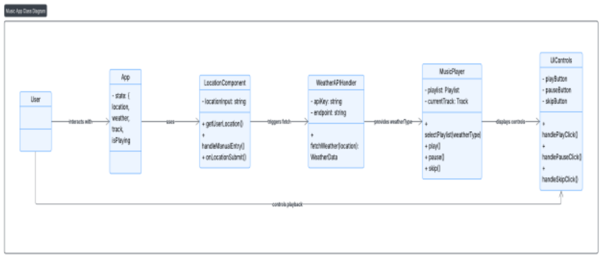
-This shows what the system does from the user's perspective.

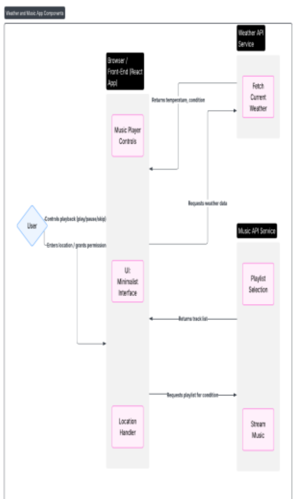
-Actors: user, weather provider API, database.

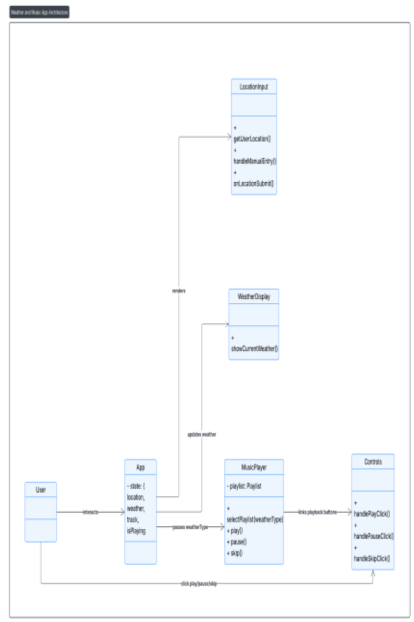
-Core interactions: enter location, fetch weather, generate playlist, play music.

-External integrations shown as separate systems

1. Front-end Architecture







UML #2: Component Diagram

-This shows the main building blocks and how they connect.

-Frontend: React components for location entry, weather display, player, and theme.

-Backend: Node controllers and services.

-External API: Weather provider.

-Emphasizes data flow through HTTPS and internal service calls.

The front end is implemented with React, which makes it component-based in structure. The application allows starting with a basic page setup design that goes with a minimalistic UI to keep the focus on the content and not on the interface. Key UI components include:

- **Location Entry & Selector:** Allows users to either grant location or manually input the city.
- **Weather Display Widget:** It shows real-time weather conditions fetched from third-party weather API.
- **Audio Player Component:** Provides playback controls for start, pause, and skip.

Everything communicates through React state management and context to make sure that if one thing changes, location, weather, then the UI will also change.

4. External Integrations

The system integrates the Weather API, OpenWeather or other, to fetch real-time data depending on user location permissions or manual input. The music can be provided through premade playlists that correspond to different weather conditions such as sunny, windy, rainy, and cloudy.

Conclusion

This architecture uses React for UI for the frontend and we use the external weather API from Open Weather Map for real-time weather and integration of music. This sets the stage for sprint 2 in which we build the backend logic and connect to the Postgres Database